GROUP BY

SELECT C1, C2, C3 FROM Sample_table WHERE

C1 = "T" GROUP BY C1, C2, C3

Result

Data grouped into categories C1, C2, C3

SELECT T_id, T_status from Transaction

Group By T_id, T_status

T_id	T_status
1	Y
2	N
3	Y
4	N
5	Y

Y = Success

N = X

successful Category

T_id	T_status
1	Y
3	Y
5	Y
2	N
4	N

FROM → WHERE → GROUP BY → SELECT → HAVING

→ Group By statement groups rows with same values in a particular column.

⇒ use Bank;

select * from Card;

---

select

is_blocked from card;

Group by

is_blocked;

same results

SELECT DISTINCT is_blocked FROM card;

unique values

Thus, we can find distinct element in the table using 2 ways –

1) SELECT column_name FROM table_name GROUP BY column_name.

2) SELECT DISTINCT column_name FROM table_name

Both queries will have same op.

Aggregate Functions

1) COUNT ()
2) SUM ()
3) AVG ()
4) MIN ()
5) MAX ()

C1, C2 → (R6 ... R10)
↓
(R1, ..., R5)

SELECT col1, col2 FROM table_name
GROUP BY Col1, Col2

R1	Y	1
R2	N	2
R3	Y	3
R4	N	4
R5	Y	5
R6	N	6

Y	N
R1 R3 R5	R2 R4 R6

Then we want to count

count = 3

SELECT col1, col2,
COUNT (Col1, Col2)
FROM Sample_table
GROUP BY Col1, Col2

Account ID	A_Type	A_Balance
1	A	100
2	B	200
3	B	300

- UPPER () and LOWER () : can't be used with group-by.

```
SELECT deptid, SUM (salary)
FROM employees
Group by DeptID
```

⇓

group of aggregate function because it performs addition on a group of employees worked in a particular department.

eg SELECT COUNT(*) FROM Product where min_age =10

⇓

We'll count total entries where

```
SELECT is_blocked, count (*)
FROM
    CARD
GROUP BY
    Is_blocked)
```

Best Practice count (is_blocked) will give us count for each Category

eg SELECT city, COUNT (CITY) AS cust_num
FROM CUSTOMER
Group by city

Group By will come first then Order By

SELECT city, count (city) As Number
FROM customer
Group By city
order by Number Desc.

Having

SELECT col1
FROM table_name
WHERE colname = '1'
GROUP BY col1
HAVING key = value

key: Column

Also, conditioning but applies on result dataset of Group by.

eg:

```
SELECT   T_id , T_status  COUNT (T_status)
FROM     Transactions
GROUP BY T_id , T_status
HAVING   T_status = "Y" ;
```

will give for only this status

Q:

```
SELECT   COUNT (*)
FROM     Products_table
WHERE    Product_quantity > 5
```

Counts no. of rows